Chromosomes II

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Crossing over exchanges genetic information. It occurs when the chromosomes in homologous pairs cross over each other, breaking pieces off, which they then exchange with each other. At this point, the chromosomes are in their duplicated state, so many different combinations can occur.

1 Carry out the following exercise, working in small groups.

Step 1: Decide on two different colours for each of the homologous pairs shown at the bottom of the page (for example red/white for one pair and blue/yellow for another). Agree on a sequence for how your group will colour the four sets. Each student in the group should end up with homologous groups that look alike.

Step 2: Working alone, cut out your homologous pairs. Align them and randomly cut part of one arm away. Swap the pieces around, rejoining the genes from one (for example red/white) with the other (for example blue/yellow). Repeat the process several times, including all four arms, then return to your group and compare crossovers.

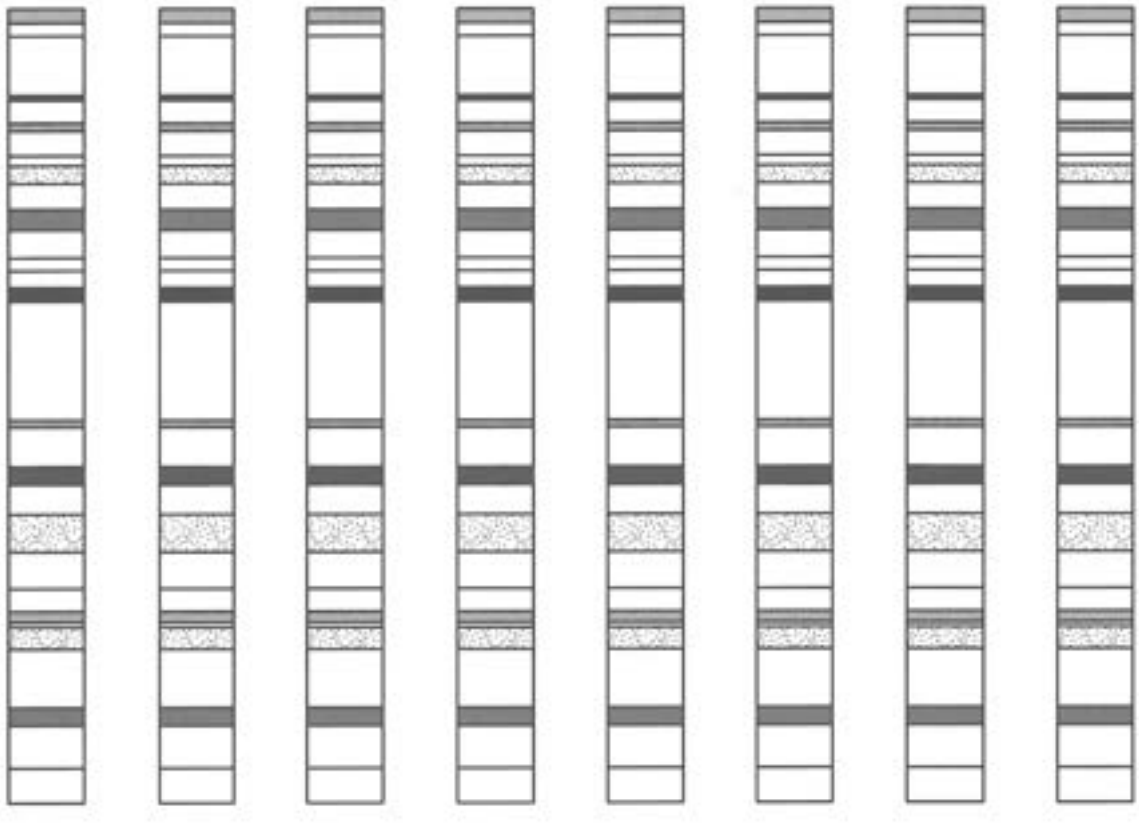
2 Did all crossovers produce identical outcomes?

3 Explain your answer to Question 2.

4 Look at the results of other students. What do they tell you about 'gene-shuffling'?

In reality, crossovers occur again and again, sometimes returning part of the genetic material to the original chromatid.

5 Why do you think that some family members resemble each other more closely than others?



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